

3.3 CLASSIFICATION AND EFFECTS OF TAXES

MARKING SCHEME

1990/CE/II/47 D	2000/CE/II/43 B	1990/CE/II/59 B	1999/CE/II/42 D	2005/CE/II/42 B (28%)
1990/CE/II/55 B	2002/CE/II/43 D (56%)	1991/CE/II/37 B	1999/CE/II/43 A	2005/CE/II/43 B (44%)
1991/CE/II/37 B	2003/CE/II/38 A (31%)	1991/CE/II/39 B	1999/CE/II/44 C	2006/CE/II/39 B (64%)
1992/CE/II/36 A	2004/CE/II/41 D (69%)	1992/CE/II/43 B	2001/CE/II/35 A	2006/CE/II/42 A (56%)
1993/CE/II/45 C	2004/CE/II/42 D (33%)	1993/CE/II/46 A	2001/CE/II/36 B	2007/CE/II/42 A (78%)
1994/CE/II/47 D	2005/CE/II/39 C (66%)	1993/CE/II/47 B	2001/CE/II/39 D	2007/CE/II/43 B (83%)
1995/CE/II/45 C	2005/CE/II/40 A (35%)	1993/CE/II/48 C	2002/CE/II/40 A (73%)	2008/CE/II/41 B (80%)
1995/CE/II/46 D	2007/CE/II/40 D (70%)	1994/CE/II/48 B	2002/CE/II/42 B (46%)	2008/CE/II/44 B (30%)
1995/CE/II/48 B	2008/CE/II/43 D (61%)	1996/CE/II/43 C	2003/CE/II/39 B (38%)	2009/CE/II/42 C (59%)
1996/CE/II/41 A	2009/CE/II/42 C (59%)	1996/CE/II/45 D	2003/CE/II/40 C (72%)	2009/CE/II/44 C (59%)
1996/CE/II/44 B	2009/CE/II/43 D (34%)	1997/CE/II/45 A	2003/CE/II/41 A (59%)	2010/CE/II/42 D (51%)
1997/CE/II/40 C	2010/CE/II/41 C (74%)	1998/CE/II/37 C	2004/CE/II/41 D (69%)	2013/DSE/I/36 C (80%)
1998/CE/II/39 D	2014/DSE/I/36 D (52%)	1998/CE/II/42 B	2004/CE/II/43 D (63%)	2016/DSE/I/36 C (61%)
2000/CE/II/41 A	2017/DSE/I/35 A (57%)	1999/CE/II/40 B	2005/CE/II/41 B (53%)	2018/DSE/I/35 B (49%)
2020/DSE/I/36 B	2020/DSE/I/38 A	2020/DSE/I/41 C	2023/DSE/I/23 A	2023/DSE/I/36 C

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/CE/I/5(b)(i)

Policy A widens the wealth gap (1)
since the poor pay more in proportion to their income on consumption than the rich (2)
compared with the rich, the poor pay a larger portion of their income as tax (2)

(max: 4)

Policy B widens the wealth gap

benefits the rich as their tax burden is lowered and income after tax ↑ (1)

no effect on the poor because they are outside the tax net or not paying at the standard rate (2)

(max: 4)

1992/CE/1/5(c)(i)

- raise direct taxes
(disadv: work disincentive / investment disincentive)
 - raise indirect taxes
(disadv: inflationary pressure / more uneven distribution of real income)
 - use past fiscal reserves
(disadv: reduce the reserves for contingent purposes / interest return from the investment of the reserves is forsaken / inflationary pressure)
 - borrowing / issue government bonds
(disadv: the economy has to bear interest cost / burden will be borne by the future generations when the principal is repaid)
 - invite private sector participation
(disadv: reduce government say in the related services)
 - cut other government expenditures
(disadv: less public services in other aspects)
 - raise other sources of government revenue, e.g. privatization / sale of the public assets, increase fees
(disadv: more burden on the users of the public services)
- (3@, max: 9)
- (Mark the **FIRST THREE** points only.)

1993/CE/1/5

- (b) (i) It is a proportional tax, because (1)
a fixed portion of the rental income is taxed (2)

(d) For:

- increase government revenue
 - discourage speculation in the property market
 - widens the tax base / sources of tax revenue
 - a tax on earning / ability-to-pay
- (2@, max: 2)
- (Mark the **FIRST** point only.)

Against:

- high administration cost
 - against laissez-faire policy / freedom to buy and sell
 - against simple taxation principle
 - distort resource allocation
 - not fair if only apply to the property market
- (2@, max: 2)
- (Mark the **FIRST** point only.)

1994/CE/1/10(b)

- (i) First progressive (because the tax rate first rises as income rises), but (2)
then proportional (because the tax rate later becomes a flat rate as income rises). (2)

- (ii) (I) - a greater amount of tax allowances (raised from \$119 000 to \$146 000) / more low-income families fall out of the tax net
- families have to earn more than \$483 000 (\$391 500 in 1992/93) before paying tax at the standard rate
- families with an annual income of more than \$119 000 and up to \$483 000 pay less tax./ average tax rate ↓
- less progressivity before reaching the standard rate
- (2@, max: 4)

(Mark the **FIRST TWO** points only.)

- (ii) (II) The new tax structure, because (2)
(to compare with the old tax structure,) the low-income and sandwich class families earning less than and up to \$483 000 a year pay a lower tax rate, but (2)
those earning more than \$483 000 a year pay the same tax rate. (2)
(Exact figures are **NOT** required in candidates' answers.)

- (iii) Workers would have a higher incentive to work, and (2)
thus earn higher income (i.e. a larger tax base). (1)

1995/CE/I/5

Progressive tax : the tax rate ↑ as income ↑ (2)
e.g., salaries tax / profits tax (fiscal year 2018/19 onwards) (1)
(Mark the **FIRST** example only.)

Proportional tax: the tax rate is constant as income ↑ (2)
e.g., salaries tax for those who pay at the standard rate / property tax (1)
(Mark the **FIRST** example only.)

1996/CE/I/10(b)

Regressive tax, because (1)
the low-income group spends a greater proportion of income on clothing than the high-income group, and therefore (2)
the low-income group pays a larger proportion of income as tax than the high-income group. (2)
(max: 3)

OR

the same amount of tax on clothing represents a greater proportion of income for the low-income group. (3)

1996/CE/I/12(d)

Increased progressivity means the tax burden of taxpayers with higher salary income increased at a faster rate than the low-income group / unemployed. (2)

The provision of unemployment benefits will benefit the low-income group / unemployed but not the high-income group. (1)
∴ More even income distribution. (1)

1997/CE/I/9(d)(ii)

The low-income group spend most of their income on purchasing goods and services for their living, (1)
but the high-income group spend only a small amount of their income for their living. (1)

OR

The low-income group pay a larger portion of their income than the high-income group on the general sales tax. (2)
(max: 2)
⇒ More uneven income distribution (1)

1998/CE/I/5

Proposal A: the tax payment takes a decreasing proportion of taxable income as taxable income rises. (2)
Proposal B: the tax payment takes a increasing proportion of taxable income as taxable income rises. (2)
Proposal C: the tax payment takes the same proportion of taxable income as taxable income rises. (2)
∴ Proposal B is the most effective. (1)

1998/CE/I/7(b)

During inflation, the nominal income of people increases. (1)
With no change in tax allowance, more people are put into the tax net or into higher tax brackets. (2)

1999/CE/I/8

Tax to income ratio

Average tax rates are:
 $\$1\,000 / \$10\,000 = 10\%$; $\$1\,900 / \$20\,000 = 9.5\%$; $\$2\,700 / \$30\,000 = 9\%$ (2)

OR

Marginal tax rates are:
 $\$ (1\,900 - 1\,000) / \$ (20\,000 - 10\,000) = 9\%$; $\$ (2\,700 - 1\,900) / \$ (30\,000 - 20\,000) = 8\%$ (2)

The conclusion is: not a progressive tax, because
the marginal tax rate decreases / the average tax rate decreases / the tax amount increase less proportionately than the taxable income. (3)

2000/CE/I/8

- Correct comparison of the amount of money rebated to the high-income group and the low-income group: (3)
e.g. 1 High-income group taxpayers are rebated a larger amount of money than low-income taxpayers.
e.g. 2 The government rebated money to the taxpayers but no money to the poor who were non-taxpayers.
It results in a wider wealth gap. (1)

2000/CE/I/9(a)

- (i) A regressive tax is a tax which tax rate decreases as income increases. (2)
Consumers pay the same amount of gasoline tax for purchasing the same amount of gasoline.
⇒ In proportion to individual's income, the richer pay at a lower tax rate while the poorer pay at a higher tax rate.
∴ Gasoline tax is a regressive tax. (3)
- (ii) - to reallocate resources: use more public transport but less private transport (2)
- a source of tax revenue (2)

2001/CE/I/7

- (a) Indirect tax because the tax burden can be shifted onto others (the buyers). (2)
- (b) The poor consumes a larger proportion of their income. (1)
OR
Same amount of tax for the same good paid by people with different incomes (1)
∴ A higher proportion of income of the low-income group is taxed. (1)
∴ The general sales tax increases the income inequality. (1)
(Remark: Mere mentioning of the conclusion without elaboration: zero marks)

2001/CE/I/11(b)

Demand side:

Cost of transaction ↓ ⇒ incentive to buy property ↑ and number of transaction ↑ (1)

Supply side:

Volume / value of business of the construction industry ↑ ⇒ expansion in the production of the construction industry and the derived demand for construction workers ↑ (1)

- (i) investment ↑ (1)
(ii) employment opportunities ↑ (1)
(Remark: Mere mentioning of the conclusion without elaboration: zero marks)

2002/CE/I/12(c)

- (i) Rates is an indirect tax because (1)
the tax burden can be shifted by the landlord onto the tenants. (1)
- (ii) The profits tax is a proportional tax because (1)
its effective tax rate is constant / the tax amount is proportional to the taxable income. (2)

2003/CE/I/10(d)

- No, because (1)
as his income increases, the tax rate is not fixed / the tax rate decreases. (2)

2004/CE/I/10(c)

- Regressive tax, because (1)
same tax payment relative to different incomes (1)
⇒ decreasing tax rate as income rises (1)

2006/CE/I/11(b)(i)(II)

- The tax is an indirect tax because (1)
the tax burden can be shifted from the producers to the consumers. (1)

2008/CE/I/8

- (a) Regressive tax, because (1)
all red wine buyers pay the same amount of tax for purchasing the same bottle of red wine (1)
higher-income earners pay a less than proportionate amount of tax / the tax rate decreases as the income increases. (2)
- (b) Indirect tax, because (1)
the tax burden can be shifted by the red wine sellers onto the consumers. (1)

2009/CE/I/11

- (a) A reduction in the profits tax rate may boost more business investment and increase profits. (2)
No, because (1)
the increase in government tax revenue resulting from the increase in profit earnings could be big enough to offset the decrease in government tax revenue resulting from the reduction in the tax rate. (1)
- (b) Indirect tax, because (1)
the tax burden can be shifted from the landlords onto the tenants. (1)

2012/DSE/II/4

- (a) Wider (1)
- (b) Lower, because (1)
higher income group has a lower post-tax income while the income of lower income group remains unchanged. (2)

2013/DSE/II/8

Government should increase the income tax rate to reduce the budget deficit

- if the increase in tax revenue due to the rise in tax rate outweighs the reduction in tax payment due to the reduction in the tax base/taxable income (e.g., a fall in the number of tax payers as a result of disincentive effects to labour). (4)
- because increase in tax rate is fairer as income tax is often proportional (or even progressive) and this helps equalize income. (2)
- if there is an inflationary (output) gap as an increase in income tax reduces output and relieves the inflationary pressure. (2)

Government should NOT increase the income tax rate to reduce the budget deficit

- if the increase in tax revenue due to the rise in tax rate falls short of the reduction in tax payment due to the reduction in the tax base/taxable income (e.g., a fall taxable income as a result of reduction in aggregate output). (4)
 - because an increase in income tax rate will result in a greater disincentive to work and more people may quit the job and rely on welfare programmes and this further worsens the budget deficits. (2)
 - as an increase in income tax rate reduces the aggregate output, which may result in unemployment / deflationary (income) gap. (2)
- (max: 4)

2014/DSE/II/6(b)

- Direct tax, because (1)
the tax burden cannot be shifted by these tax-payers to someone else. (1)

2014/DSE/II/7(a)

- negative output effect (fall in GDP) due to a reduction in government expenditure and thus aggregate demand
- negative output effect (fall in GDP) due to a rise in sales/income tax, which would lower consumption/investment demand and thus aggregate demand
- negative (disincentive) effect on labour supply and thus on employment (and output) as a result of higher income taxes

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2015/DSE/II/11(a)

- (i) The percentage change in direct tax revenue and percentage change in nominal GDP tend to move in the same direction. (1)
 Fluctuation of the former is greater than that of the latter. (1)
- (ii) Growth in the economy (increases in GDP) would be accompanied by growth in salary and profit incomes, which would in turn generate higher revenue from taxes on such incomes. (1)
 Progressive taxation may be one explanation for why direct tax revenue fluctuated more than GDP. (1)

2017/DSE/II/13(b)

- (ii) Proposal A's effect on income distribution:
 Salaries tax payers (higher-income group) would pay more taxes while non-tax-payers (lower-income group) would not be affected. (2)
OR
 High-income tax payers pay more, low-income tax payers pay less. (2)
 The after-tax income inequality would thus be reduced. (1)
 (max: 3)

Proposal B's effect on income distribution:
 There will be no extra effect on current tax payers, so the income inequality is unaffected. (2)
 Reduction of fiscal reserves may result in a tax increase in the future, so future tax payers (mainly the younger portion of the population) may have to pay more taxes, thus *increasing income inequality between generations*. (2)
 (max: 2)

- (iii) Proposal A's effect on other variables:
 Price level would drop as AD drops (while there is no such an effect generated by proposal B). (2)
 Unemployment may increase as AD drops (while there is no such an effect generated by proposal B). (2)
 (max: 2)

Proposal B's effect on other variables:
 A worsening of the government's budgetary position (due to a drain on fiscal reserves) may imply tax increase in the future. (No such effect arises under proposal A.) (2)
 Business confidence may be weakened. Expecting tax increases in the future (due to the problem of fiscal sustainability), which would reduce their after-tax profits, business firms may choose to cut down their investment. (No such effect arises under proposal A.) (2)
 (max: 2)

2018/DSE/II/6(a)

- No, because (1)
 the reduction in the profits-tax rate may induce firms to increase their investment expenditure, so they may end up earning higher profits (tax base). The increase in profits tax revenue due to the increase in profits (tax base) may be greater than the decrease in profit tax revenue due to the reduction in tax rate. (3)

2023/DSE/II/12d

(d) Marks for effective communication (EC: max 2 marks)

Marks	Performance
2	● Support arguments with the source/data and appropriate economic theories. ● Present relevant material. ● Present well-organised and coherent answers without repetition of ideas. ● Use language that expresses ideas clearly and fluently with appropriate use of words/terms/symbols.
1	● Present arguments with some support of the source/data and economic theories. ● Present some irrelevant material. ● Present answers in a less organised way with some repetition. ● Use language that conveys a clear message with some inappropriate use of words/terms/symbols.
0	● Present arguments with no support of the source/data and economic theories. ● Present material unrelated to the gist of the question. ● Present inconsistent arguments. ● Express limited ideas with inappropriate use of words/terms/symbols.

2024/DSE/II/12(d)

- (d) (i) Direct tax, because the tax burden cannot be shifted by the tax-payer (Neymar) to someone else. (2)
- (ii) Proportional tax, because the average tax rate remains unchanged (20%) when the taxable income changes. (2)

2025/DSE/II/9(Cross Topic)

- (a) It is a regressive tax, because (1)
the same amount of tax needs to be paid when staying in a hotel room of the same (1)
price regardless of income. (1)
Therefore, the average tax rate decreases as income increases. (1)
- (b) The proposed HAT is an indirect tax. As the tax rate is adjusted from 0% to 3%, the (2)
indirect tax revenue from the HAT will increase (from zero to some positive
number). (2)
- The re-imposition of the HAT may marginally reduce the profits of the hotels and (2)
guesthouses, reducing the direct tax revenue from profits tax.
- (c) (Complex) division of labour (1)
Reasons for increasing labour productivity:
 - practice makes perfect
 - assigning the most suitable person for each task
 - saving time in training
 - saving time in job rotation
 - mechanisation is stimulated
} 1@
max: 3
- [Mark the FIRST THREE points only.]

(d) Advantage:

- easier to maintain a stable team of staff as income is more stable
- lower cost of measuring outputs

[Mark the FIRST point only.]

}^{1@}
max: 1

Disadvantage:

- higher cost of monitoring against shirking
- lower labour productivity due to lower working incentive

[Mark the FIRST point only.]

}^{1@}
max: 1

- (e) No. Some people may have a lower time cost during public holidays than on other days. If the saving in time cost is greater than the additional money cost (due to a higher room rate), the opportunity cost of staying in a hotel in Hong Kong would be lower.

(3)